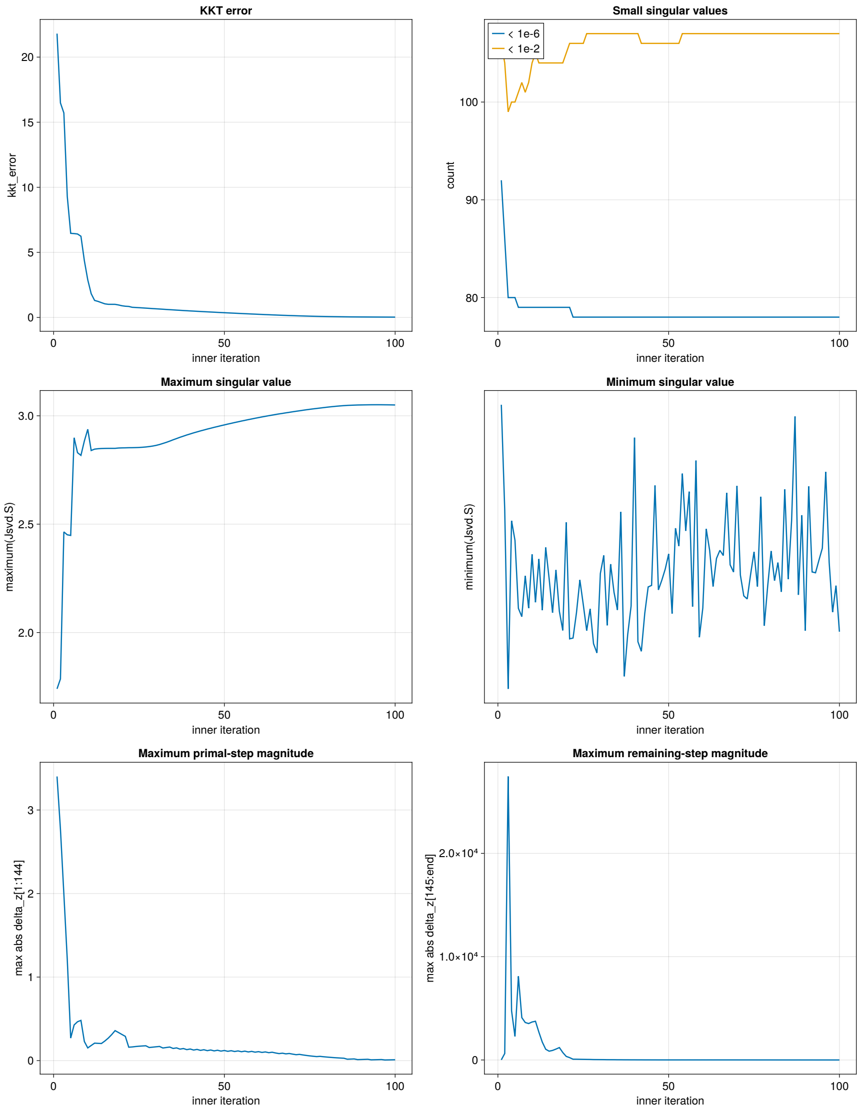
# Solver Diagnostics

Recorded 100 inner iterations; reporting 100 rows (cap: 1000).

## Summary statistics

| Metric | Minimum | Maximum | Mean | Median |
|---|---:|---:|---:|---:|
| `kkt_error` | 2.488044e-02 | 2.180502e+01 | 1.315393e+00 | 3.575086e-01 |
| `count(Jsvd.S < 1e-6)` | 7.800000e+01 | 9.200000e+01 | 7.844000e+01 | 7.800000e+01 |
| `count(Jsvd.S < 1e-2)` | 9.900000e+01 | 1.070000e+02 | 1.060400e+02 | 1.070000e+02 |
| `maximum(Jsvd.S)` | 1.740234e+00 | 3.050877e+00 | 2.917352e+00 | 2.959806e+00 |
| `minimum(Jsvd.S)` | 1.589580e-17 | 5.466391e-17 | 3.198239e-17 | 3.112839e-17 |
| `maximum(abs.(delta_z[1:144]))` | 7.670062e-03 | 3.401398e+00 | 2.147264e-01 | 1.161799e-01 |
| `maximum(abs.(delta_z[145:end]))` | 6.471923e-01 | 2.744207e+04 | 7.364860e+02 | 6.263878e+00 |

# Metrics versus inner iteration



## Per-iteration diagnostics

| Iter | KKT error | $N_{<10^{-6}}$ | $N_{<10^{-2}}$ | $\sigma_{\max}$ | $\sigma_{\min}$ | $\max\lvert\delta z_{1:144}\rvert$ | $\max\lvert\delta z_{145:}\rvert$ |
|---|---|---|---|---|---|---|---|
| 1 | 2.180502e+01 | 92 | 106 | 1.740234e+00 | 5.466391e-17 | 3.401398e+00 | 1.237659e+01 |
| 2 | 1.647741e+01 | 86 | 104 | 1.786013e+00 | 4.029825e-17 | 2.768943e+00 | 6.262250e+02 |
| 3 | 1.571054e+01 | 80 | 99 | 2.464213e+00 | 1.589580e-17 | 1.994494e+00 | 2.744207e+04 |
| 4 | 9.291457e+00 | 80 | 100 | 2.450910e+00 | 3.883356e-17 | 1.209092e+00 | 4.761862e+03 |
| 5 | 6.461158e+00 | 80 | 100 | 2.447946e+00 | 3.615364e-17 | 2.696704e-01 | 2.279373e+03 |
| 6 | 6.442933e+00 | 79 | 101 | 2.898662e+00 | 2.687836e-17 | 4.268086e-01 | 8.098631e+03 |
| 7 | 6.411259e+00 | 79 | 102 | 2.829750e+00 | 2.574849e-17 | 4.652447e-01 | 4.100032e+03 |
| 8 | 6.238603e+00 | 79 | 101 | 2.817407e+00 | 3.132144e-17 | 4.815635e-01 | 3.627933e+03 |
| 9 | 4.361053e+00 | 79 | 102 | 2.883618e+00 | 2.690041e-17 | 2.287508e-01 | 3.525437e+03 |
| 10 | 2.886537e+00 | 79 | 104 | 2.937607e+00 | 3.425808e-17 | 1.518073e-01 | 3.676666e+03 |
| 11 | 1.825734e+00 | 79 | 105 | 2.839783e+00 | 2.770566e-17 | 1.809719e-01 | 3.748567e+03 |
| 12 | 1.307839e+00 | 79 | 104 | 2.846098e+00 | 3.360870e-17 | 2.091046e-01 | 2.692928e+03 |
| 13 | 1.232989e+00 | 79 | 104 | 2.847994e+00 | 2.662104e-17 | 2.065901e-01 | 1.728413e+03 |
| 14 | 1.137408e+00 | 79 | 104 | 2.849052e+00 | 3.518552e-17 | 2.047578e-01 | 1.037229e+03 |
| 15 | 1.045928e+00 | 79 | 104 | 2.849534e+00 | 3.093534e-17 | 2.346259e-01 | 8.513586e+02 |
| 16 | 1.008803e+00 | 79 | 104 | 2.849718e+00 | 2.628227e-17 | 2.692009e-01 | 9.238002e+02 |
| 17 | 1.007730e+00 | 79 | 104 | 2.849725e+00 | 3.210970e-17 | 3.109505e-01 | 1.054783e+03 |
| 18 | 1.006647e+00 | 79 | 104 | 2.849734e+00 | 2.646637e-17 | 3.589345e-01 | 1.201911e+03 |
| 19 | 9.594375e-01 | 79 | 104 | 2.851265e+00 | 2.386904e-17 | 3.363079e-01 | 7.215429e+02 |
| 20 | 8.984870e-01 | 79 | 105 | 2.851855e+00 | 3.861309e-17 | 3.130424e-01 | 3.470088e+02 |
| 21 | 8.615550e-01 | 79 | 106 | 2.852175e+00 | 2.271868e-17 | 2.904453e-01 | 2.275169e+02 |
| 22 | 8.464709e-01 | 78 | 106 | 2.852663e+00 | 2.282820e-17 | 1.605209e-01 | 7.508392e+01 |
| 23 | 7.805205e-01 | 78 | 106 | 2.852776e+00 | 2.626375e-17 | 1.635746e-01 | 7.251931e+01 |
| 24 | 7.659232e-01 | 78 | 106 | 2.853199e+00 | 3.074930e-17 | 1.685410e-01 | 6.545528e+01 |
| 25 | 7.499262e-01 | 78 | 106 | 2.853832e+00 | 2.738957e-17 | 1.722607e-01 | 5.946399e+01 |
| 26 | 7.333977e-01 | 78 | 107 | 2.854756e+00 | 2.384681e-17 | 1.751874e-01 | 5.432471e+01 |
| 27 | 7.163520e-01 | 78 | 107 | 2.856091e+00 | 2.681655e-17 | 1.776312e-01 | 4.985033e+01 |
| 28 | 6.988534e-01 | 78 | 107 | 2.857999e+00 | 2.205574e-17 | 1.567605e-01 | 3.994298e+01 |
| 29 | 6.806581e-01 | 78 | 107 | 2.860294e+00 | 2.080022e-17 | 1.611134e-01 | 3.758635e+01 |
| 30 | 6.646619e-01 | 78 | 107 | 2.863385e+00 | 3.164319e-17 | 1.654437e-01 | 3.548062e+01 |
| 31 | 6.480650e-01 | 78 | 107 | 2.867381e+00 | 3.409745e-17 | 1.696258e-01 | 3.335877e+01 |
| 32 | 6.310217e-01 | 78 | 107 | 2.872325e+00 | 2.455783e-17 | 1.510099e-01 | 2.726596e+01 |
| 33 | 6.129464e-01 | 78 | 107 | 2.877378e+00 | 3.290945e-17 | 1.568824e-01 | 2.584566e+01 |
| 34 | 5.972064e-01 | 78 | 107 | 2.883079e+00 | 2.907548e-17 | 1.623739e-01 | 2.455374e+01 |
| 35 | 5.808835e-01 | 78 | 107 | 2.889249e+00 | 2.666570e-17 | 1.455489e-01 | 2.026483e+01 |
| 36 | 5.632215e-01 | 78 | 107 | 2.894891e+00 | 4.004173e-17 | 1.522087e-01 | 1.929198e+01 |
| 37 | 5.480022e-01 | 78 | 107 | 2.900772e+00 | 1.759625e-17 | 1.374722e-01 | 1.610738e+01 |
| 38 | 5.314327e-01 | 78 | 107 | 2.906021e+00 | 2.328366e-17 | 1.446742e-01 | 1.547075e+01 |
| 39 | 5.169453e-01 | 78 | 107 | 2.911434e+00 | 2.719556e-17 | 1.313469e-01 | 1.303053e+01 |
| 40 | 5.011535e-01 | 78 | 107 | 2.916242e+00 | 5.017375e-17 | 1.388417e-01 | 1.259923e+01 |
| 41 | 4.872444e-01 | 78 | 107 | 2.921211e+00 | 2.233084e-17 | 1.265237e-01 | 1.068416e+01 |
| 42 | 4.720348e-01 | 78 | 106 | 2.925639e+00 | 2.101597e-17 | 1.341668e-01 | 1.038447e+01 |
| 43 | 4.585988e-01 | 78 | 106 | 2.930245e+00 | 2.619082e-17 | 1.225953e-01 | 8.854598e+00 |
| 44 | 4.438455e-01 | 78 | 106 | 2.934371e+00 | 2.980096e-17 | 1.302910e-01 | 8.643462e+00 |
| 45 | 4.308093e-01 | 78 | 106 | 2.938693e+00 | 3.001117e-17 | 1.192845e-01 | 7.404697e+00 |
| 46 | 4.164300e-01 | 78 | 106 | 2.942586e+00 | 4.365144e-17 | 1.269627e-01 | 7.255603e+00 |
| 47 | 4.037433e-01 | 78 | 106 | 2.946686e+00 | 2.940251e-17 | 1.163916e-01 | 6.241708e+00 |
| 48 | 3.896846e-01 | 78 | 106 | 2.950376e+00 | 3.072572e-17 | 1.239963e-01 | 6.137432e+00 |
| 49 | 3.773151e-01 | 78 | 106 | 2.954322e+00 | 3.224473e-17 | 1.137660e-01 | 5.332924e+00 |
| 50 | 3.635439e-01 | 78 | 106 | 2.957909e+00 | 3.429772e-17 | 1.212486e-01 | 5.806451e+00 |
| 51 | 3.514734e-01 | 78 | 106 | 2.961703e+00 | 2.615819e-17 | 1.112890e-01 | 5.470016e+00 |
| 52 | 3.379714e-01 | 78 | 106 | 2.965149e+00 | 3.781510e-17 | 1.186045e-01 | 5.948982e+00 |
| 53 | 3.261931e-01 | 78 | 106 | 2.968810e+00 | 3.538886e-17 | 1.088635e-01 | 5.598322e+00 |
| 54 | 3.129532e-01 | 78 | 107 | 2.972136e+00 | 4.526845e-17 | 1.159682e-01 | 6.079705e+00 |
| 55 | 3.014702e-01 | 78 | 107 | 2.975668e+00 | 3.745470e-17 | 1.064072e-01 | 5.713503e+00 |
| 56 | 2.884947e-01 | 78 | 107 | 2.978873e+00 | 4.280717e-17 | 1.132572e-01 | 6.193786e+00 |
| 57 | 2.773179e-01 | 78 | 107 | 2.982270e+00 | 2.711272e-17 | 1.038483e-01 | 5.810907e+00 |
| 58 | 2.646173e-01 | 78 | 107 | 2.985347e+00 | 4.704369e-17 | 1.103986e-01 | 6.286048e+00 |
| 59 | 2.537653e-01 | 78 | 107 | 2.988603e+00 | 2.293214e-17 | 1.011225e-01 | 5.885564e+00 |
| 60 | 2.413573e-01 | 78 | 107 | 2.991548e+00 | 2.693195e-17 | 1.073268e-01 | 6.350983e+00 |
| 61 | 2.308562e-01 | 78 | 107 | 2.994660e+00 | 3.771609e-17 | 9.817162e-02 | 5.932210e+00 |
| 62 | 2.187658e-01 | 78 | 107 | 2.997476e+00 | 3.480191e-17 | 1.039822e-01 | 6.382804e+00 |

| Iter | KKT error | $N_{<10^{-6}}$ | $N_{<10^{-2}}$ | $\sigma_{\max}$ | $\sigma_{\min}$ | $\max\lvert\delta z_{1:144}\rvert$ | $\max\lvert\delta z_{145:}\rvert$ |
|---|---|---|---|---|---|---|---|
| 63 | 2.086497e-01 | 78 | 107 | 3.000449e+00 | 2.986640e-17 | 9.494273e-02 | 5.945355e+00 |
| 64 | 1.969089e-01 | 78 | 107 | 3.003145e+00 | 3.368893e-17 | 1.003109e-01 | 6.375541e+00 |
| 65 | 1.872224e-01 | 78 | 107 | 3.005997e+00 | 3.479817e-17 | 9.138799e-02 | 5.919404e+00 |
| 66 | 1.758696e-01 | 78 | 107 | 3.008593e+00 | 3.409007e-17 | 8.386541e-02 | 5.513954e+00 |
| 67 | 1.657477e-01 | 78 | 107 | 3.010985e+00 | 4.264877e-17 | 8.872284e-02 | 5.908362e+00 |
| 68 | 1.571536e-01 | 78 | 107 | 3.013547e+00 | 3.281958e-17 | 8.096400e-02 | 5.480754e+00 |
| 69 | 1.472792e-01 | 78 | 107 | 3.015914e+00 | 3.183742e-17 | 8.529363e-02 | 5.844297e+00 |
| 70 | 1.391713e-01 | 78 | 107 | 3.018457e+00 | 4.359458e-17 | 7.753441e-02 | 5.395789e+00 |
| 71 | 1.296412e-01 | 78 | 107 | 3.020820e+00 | 3.142536e-17 | 7.092180e-02 | 4.996189e+00 |
| 72 | 1.211329e-01 | 78 | 107 | 3.023035e+00 | 2.858255e-17 | 7.464063e-02 | 5.313084e+00 |
| 73 | 1.141862e-01 | 78 | 107 | 3.025444e+00 | 2.816250e-17 | 6.780830e-02 | 4.891599e+00 |
| 74 | 1.059542e-01 | 78 | 107 | 3.027705e+00 | 3.153302e-17 | 6.193885e-02 | 4.515226e+00 |
| 75 | 9.858286e-02 | 78 | 107 | 3.029841e+00 | 3.459296e-17 | 5.676073e-02 | 4.177390e+00 |
| 76 | 9.192178e-02 | 78 | 107 | 3.031865e+00 | 2.985336e-17 | 5.216429e-02 | 3.872714e+00 |
| 77 | 8.586556e-02 | 78 | 107 | 3.033786e+00 | 4.211298e-17 | 4.806212e-02 | 3.596796e+00 |
| 78 | 8.033208e-02 | 78 | 107 | 3.035610e+00 | 2.450133e-17 | 5.085515e-02 | 3.832038e+00 |
| 79 | 7.585242e-02 | 78 | 107 | 3.037604e+00 | 2.994518e-17 | 4.647265e-02 | 3.533006e+00 |
| 80 | 7.043135e-02 | 78 | 107 | 3.039472e+00 | 3.469532e-17 | 4.261905e-02 | 3.262879e+00 |
| 81 | 6.550072e-02 | 78 | 107 | 3.041214e+00 | 3.066114e-17 | 3.916883e-02 | 3.018055e+00 |
| 82 | 6.100008e-02 | 78 | 107 | 3.042828e+00 | 3.312490e-17 | 3.606701e-02 | 2.795475e+00 |
| 83 | 5.687742e-02 | 78 | 107 | 3.044312e+00 | 2.913263e-17 | 3.326825e-02 | 2.592556e+00 |
| 84 | 5.309004e-02 | 78 | 107 | 3.045662e+00 | 4.313027e-17 | 3.073467e-02 | 2.407098e+00 |
| 85 | 4.960212e-02 | 78 | 107 | 3.046871e+00 | 3.086339e-17 | 2.843432e-02 | 2.237213e+00 |
| 86 | 4.638316e-02 | 78 | 107 | 3.047936e+00 | 3.885473e-17 | 1.679180e-02 | 1.328461e+00 |
| 87 | 4.296469e-02 | 78 | 107 | 3.048512e+00 | 5.307777e-17 | 1.843610e-02 | 1.458918e+00 |
| 88 | 4.113640e-02 | 78 | 107 | 3.049112e+00 | 2.872532e-17 | 2.003896e-02 | 1.590451e+00 |
| 89 | 3.928336e-02 | 78 | 107 | 3.049706e+00 | 3.959059e-17 | 1.195430e-02 | 9.528665e-01 |
| 90 | 3.727156e-02 | 78 | 107 | 3.050010e+00 | 2.382475e-17 | 1.324933e-02 | 1.056361e+00 |
| 91 | 3.590575e-02 | 78 | 107 | 3.050312e+00 | 4.353718e-17 | 1.456737e-02 | 1.163911e+00 |
| 92 | 3.446587e-02 | 78 | 107 | 3.050584e+00 | 3.184690e-17 | 1.590050e-02 | 1.273427e+00 |
| 93 | 3.299828e-02 | 78 | 107 | 3.050793e+00 | 3.173400e-17 | 9.517662e-03 | 7.648308e-01 |
| 94 | 3.139794e-02 | 78 | 107 | 3.050856e+00 | 3.343956e-17 | 1.057536e-02 | 8.500104e-01 |
| 95 | 3.030860e-02 | 78 | 107 | 3.050877e+00 | 3.509195e-17 | 1.166481e-02 | 9.391579e-01 |
| 96 | 2.915627e-02 | 78 | 107 | 3.050821e+00 | 4.551139e-17 | 1.277801e-02 | 1.048574e+00 |
| 97 | 2.797713e-02 | 78 | 107 | 3.050648e+00 | 3.315027e-17 | 7.670062e-03 | 6.471923e-01 |
| 98 | 2.669321e-02 | 78 | 107 | 3.050473e+00 | 2.636132e-17 | 8.541180e-03 | 7.382916e-01 |
| 99 | 2.581383e-02 | 78 | 107 | 3.050220e+00 | 2.997839e-17 | 9.447130e-03 | 8.432656e-01 |
| 100 | 2.488044e-02 | 78 | 107 | 3.049848e+00 | 2.369341e-17 | 1.038081e-02 | 9.616226e-01 |